Factorising by Grouping

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Factorising by grouping

How do I factorise expressions with a common bracket?

- Look at the expression $3x(t+4)+2(t+4)$
  - Both terms have a **common bracket**, $(t+4)$
  - The whole bracket, $(t+4)$, can be "taken out" like a common factor:
    - $(t+4)(3x+2)$
- This is like factorising $3xy+2y$ to get $y(3x+2)$
  - y represents $(t+4)$ above

How do I factorise by grouping?

- Some questions may require you to **form a common bracket yourself**
  - For example $xy+3x+5y+15$
    - The first two terms have a common factor of x
    - The second two terms have a common factor of 5
  - Factorising fully the **first pair** of terms, and the **last pair** of terms:
    - $x(y+3)+5(y+3)$
  - You can now spot a common bracket of $(y+3)$
    - $(y+3)(x+5)$
- This is called **factorising by grouping**

Does it matter what order I group in?

- You can often **rearrange terms** to factorise in a **different order**
  - Rewriting the same example, $xy+3x+5y+15$, but in a different order:
    - $xy+5y+3x+15$
    - The first pair of terms have a common factor of y
    - The second pair of terms have a common factor of 3
  - Factorising gives $y(x+5)+3(x+5)$
    - You can now spot a common bracket, this time of $(x+5)$
  - $(x+5)(y+3)$
    - This gives the **same result** as found previously

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

$3x + 5y$

Examiner Tips and Tricks

Once you have factorised something, expand it by hand to check your answer is correct.

Worked Example

Factorise $ab + 3b + 2a + 6$.

Answer:

Method 1

Notice that ab and $3b$ have a common factor of b
Notice that $2a$ and 6 have a common factor of 2

Factorise the first two terms, using b as a common factor

$$b(a + 3) + 2a + 6$$

Factorise the second two terms, using 2 as a common factor

$$b(a + 3) + 2(a + 3)$$

$(a + 3)$ is a common bracket
We can now factorise out the bracket $(a + 3)$

$$\mathbf{(a + 3)(b + 2)}$$

Method 2

Notice that ab and $2a$ have a common factor of a
Notice that $3b$ and 6 have a common factor of 3

Rewrite the expression, grouping these terms together

$$ab + 2a + 3b + 6$$

Factorise the first two terms, using a as a common factor

$$a(b + 2) + 3b + 6$$

Factorise the second two terms, using 3 as a common factor

$$a(b + 2) + 3(b + 2)$$

$(b + 2)$ is a common bracket
We can now factorise out the bracket $(b + 2)$

$$\mathbf{(b + 2)(a + 3)}$$

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